https://indico.bnl.gov/event/32005/contributions/121797/attachments/69012/118450/AuAu_gj_BSeidlitz.pdf

AuAu γ-jet

Blair Seidlitz & Shuhang Li

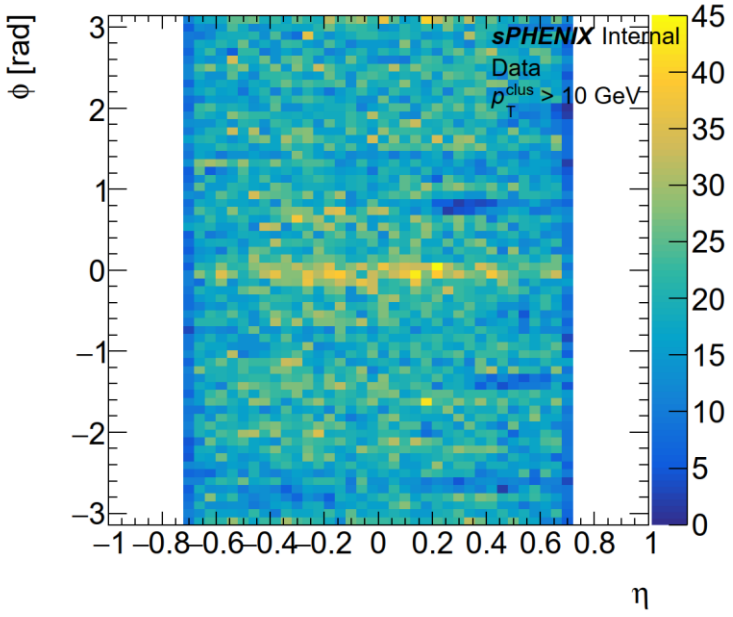
Datasets

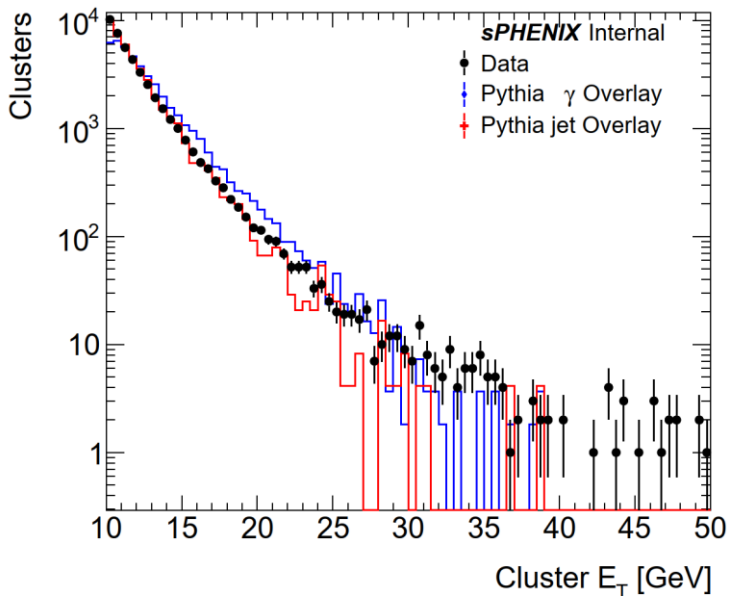
- **Data: 2025 AuAu**
  - Ran over some of the data available: need to check this
  - No Good Run List
  - Removed 1 very bad run
  - Counted raw MBD N&S > 2 in all runs analyzed, assumed a 6b cross section and see about 1 nb^{-1} lumi,
  - Keep events only if a cluster of > 10 GeV is present
- **MC: Pythia + data embedded into 24 AuAu**
  - Jet 10, 20
  - Photon 10, 20

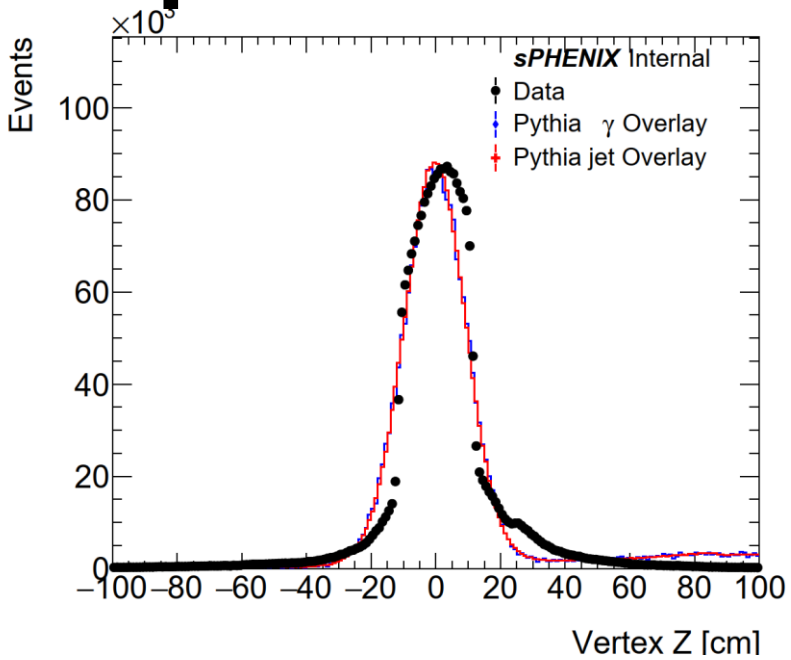
Object selection

- **Clusters**
  - Nominal clustering algorism (which includes splitting)
  - No UE subtraction
  - Using constituent sum for E, (i.e. not Ecore)
- **Jets**
  - Anti-kT R=0.4 jets made from UE subtracted towers
  - Default UE subtraction in HIJetReco
  - Focusing on jets > 8 GeV, considering going a little lower
  - All jet pTs are calibrated with the nominal pp JES

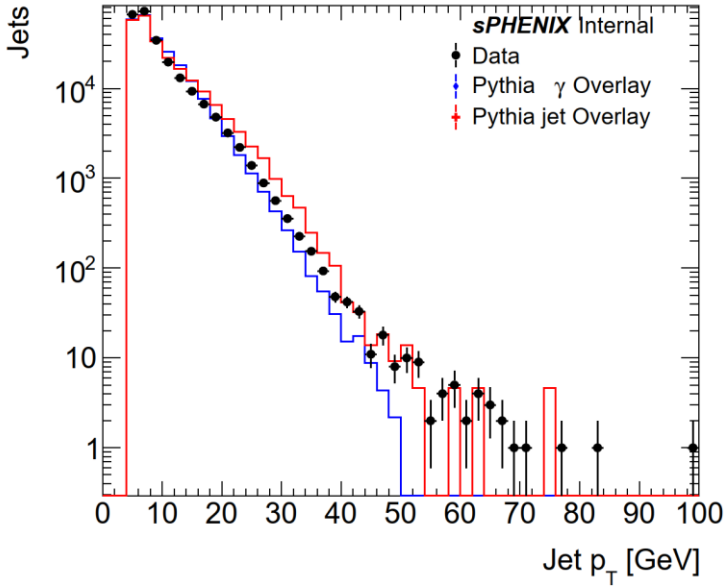
Basic comparisons data to Jet10 and photon10

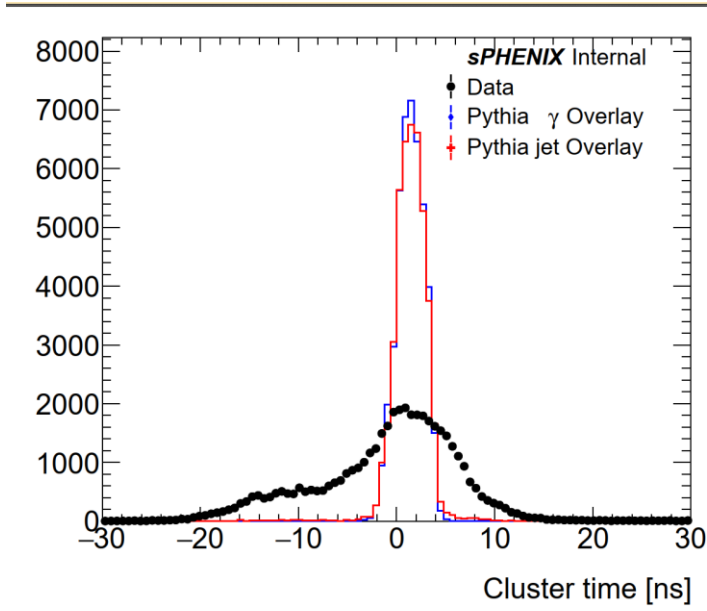






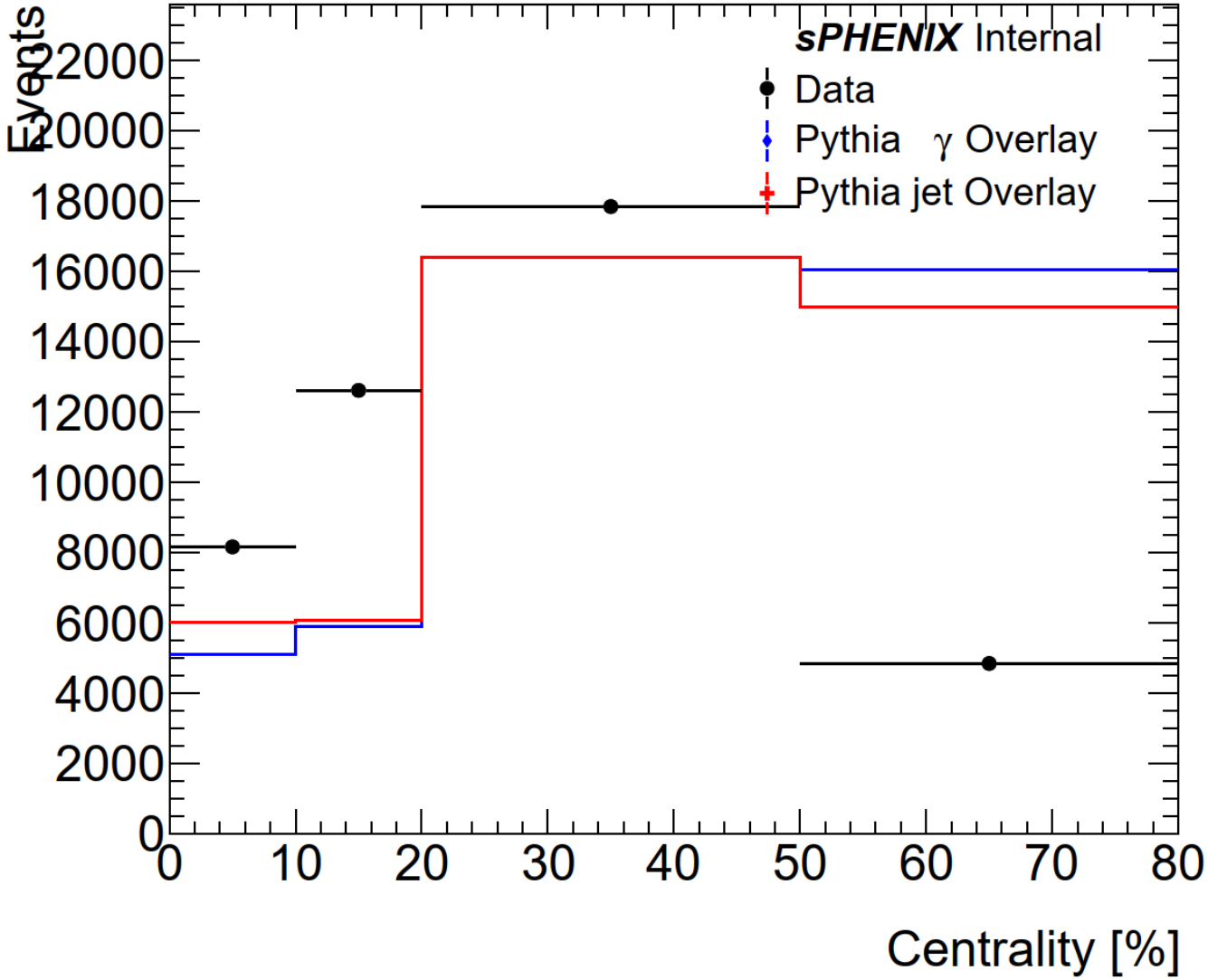
- **Left: eta-phi map of clusters**
- **Data-MC comparisons**
  - MC is normalized to counts in data
  - MC agrees suitably for initial studies
  - Some evidence of backgrounds in data





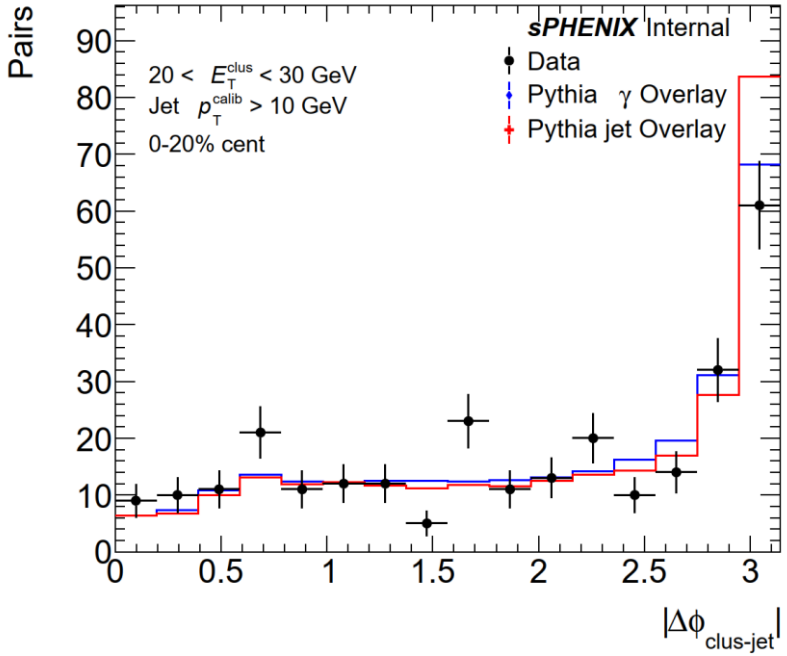
Issue with centrality in 2024 and 2025

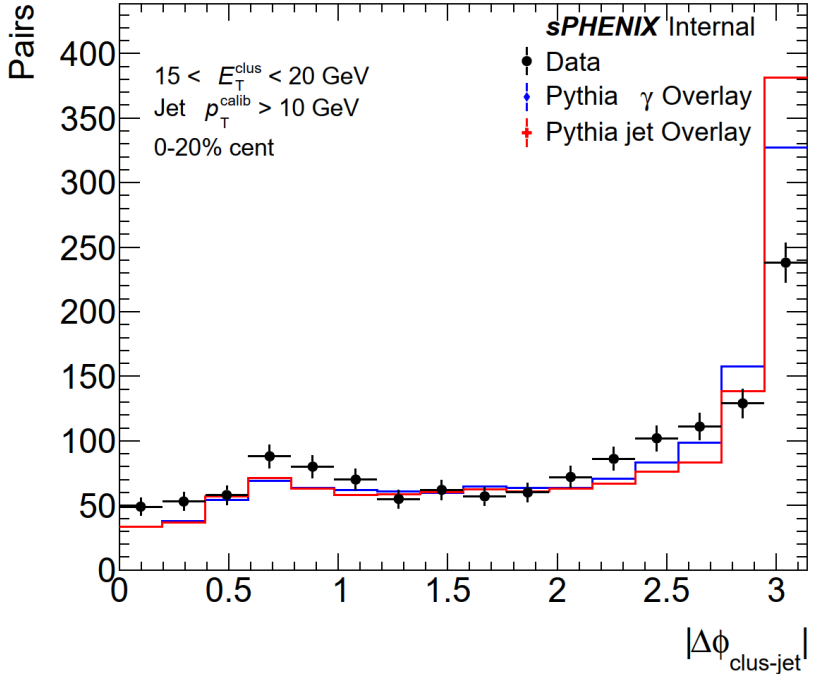
- Known issue in the centrality due to change in MBD calibration
- Expect a monotonically rising high-pt cluster yield with centrality in data
- In data overlay, one Pythia event is embedded per one MB event – should be flat
- Probably a similar thing happening in data and embedding
- Holding of on centrality reweighting

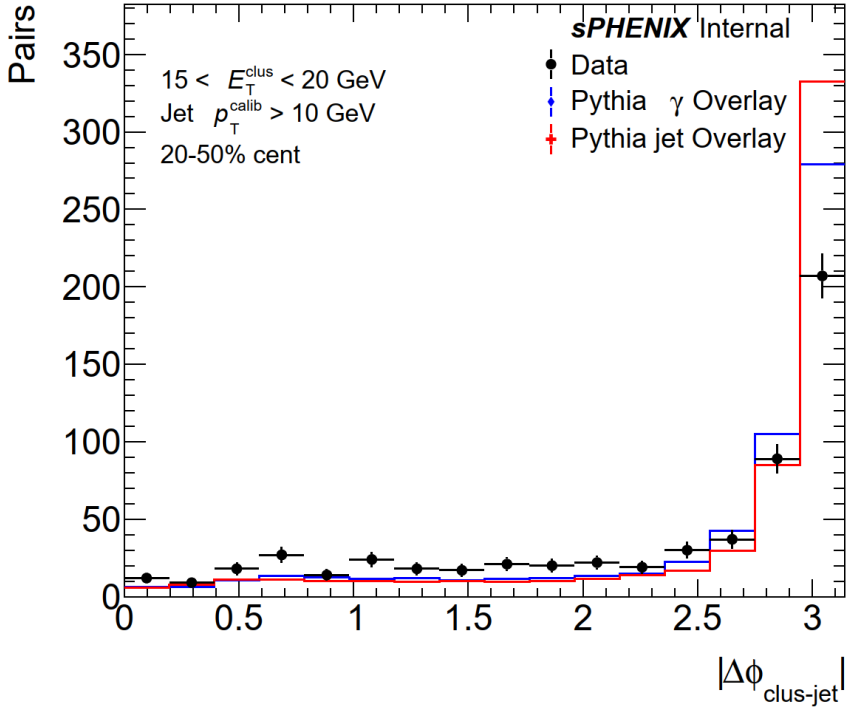


Jet-cluster Δϕ

- **Cluster-jet Δϕ**
  - Cluster-jet ΔR > 0.2
  - No selection and centrality reweighting
- **Interesting but can't read to much into it**

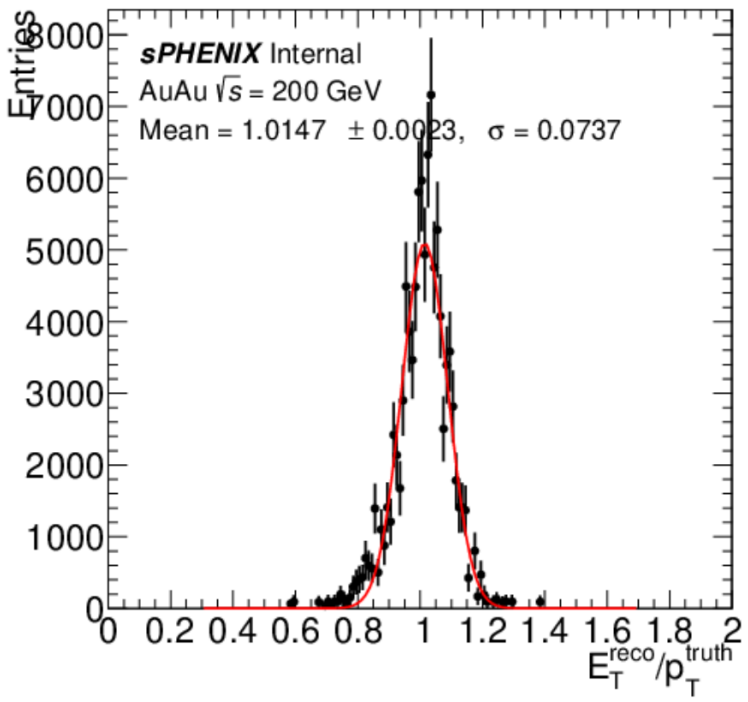


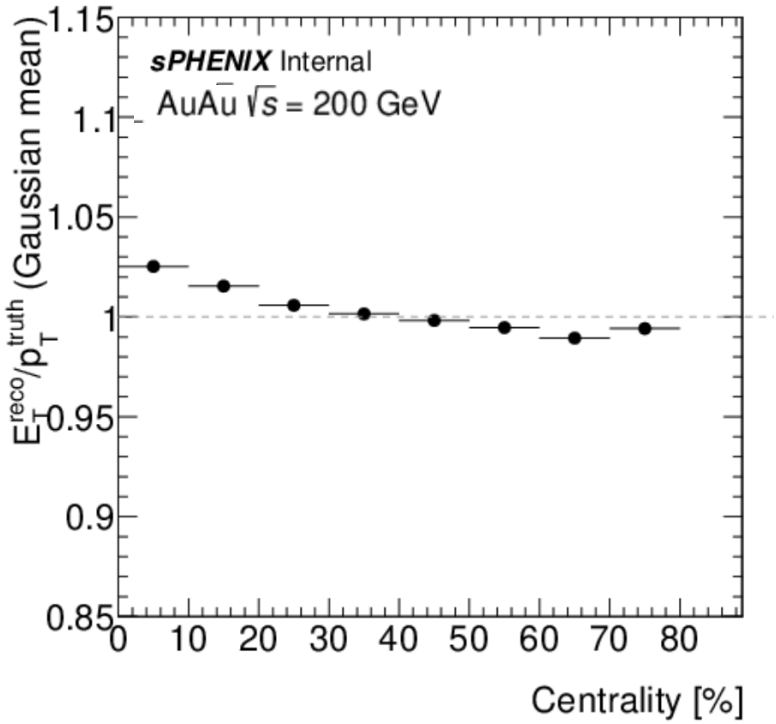


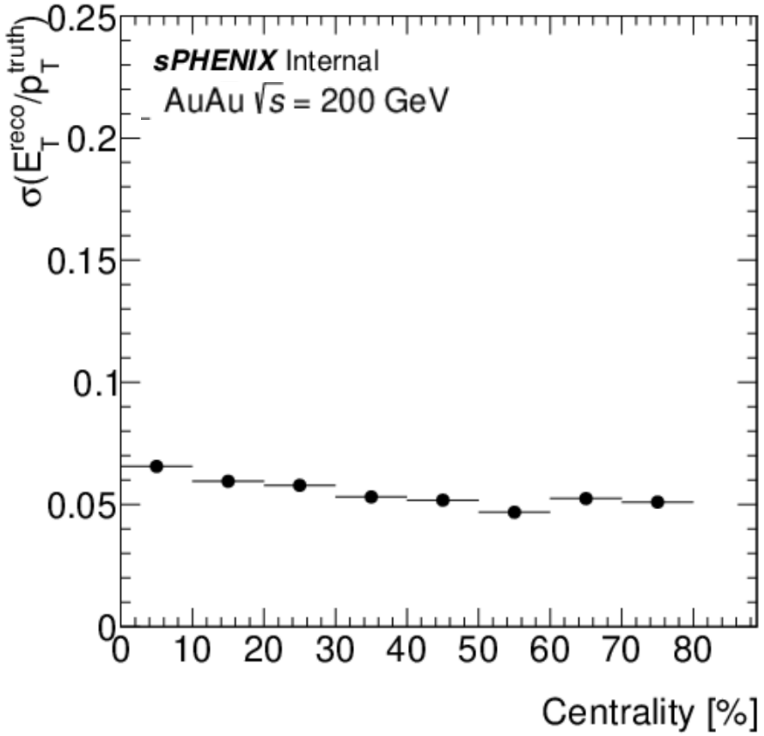


Cluster performance

- **Truth-matched photons in the signal photon10 sample**
  - Left: 0-10% response distribution for clusters > 10 GeV
  - Middle-Right: scale and resolution
- **Cluster performance experiences some degradation in central AuAu collisions but I think this is sufficient performance for now**





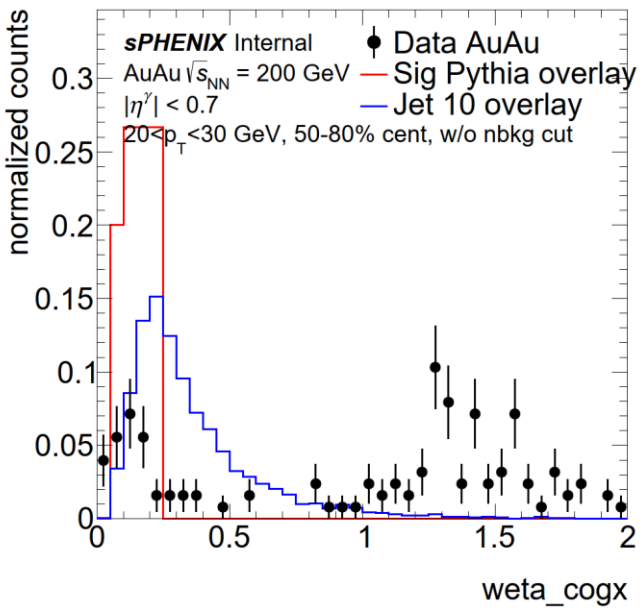


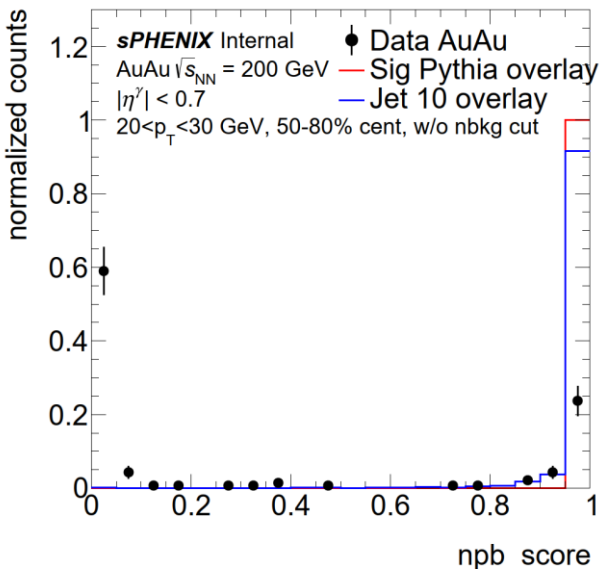
Non-physics background (NPB) in 2025 data

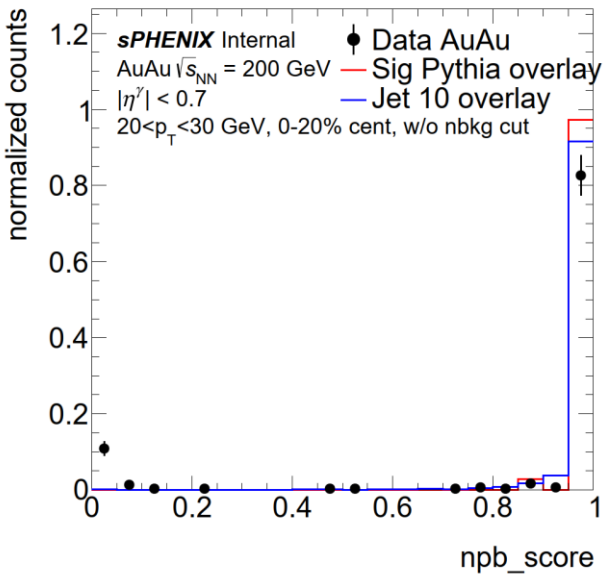
- **A large non-physics background (NPB) was observed in PPG12, most clearly in the weta_cogx shower shape variable**
  - Left: weta_cogx in peripheral AuAU

$w_{\eta(\phi)}^{\mathrm{COGX}}$	energy weighted second moment of $\eta(\phi)$ from all towers in a given cluster excluding the tower containing the center of gravity.

- **A BDT was trained in pp to remove this background which aggregated shower shape info. This seems to work in AuAu too**
  - Middle-right: NPB score in AuAU. 0 is background 1 is signal
- **Choose a cut of NPB > 0.5 (same as PPG 12)**

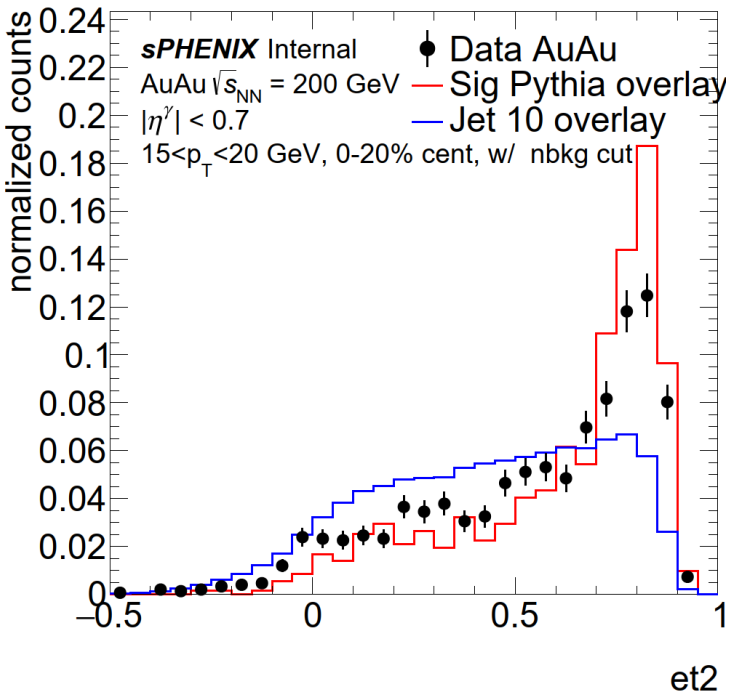


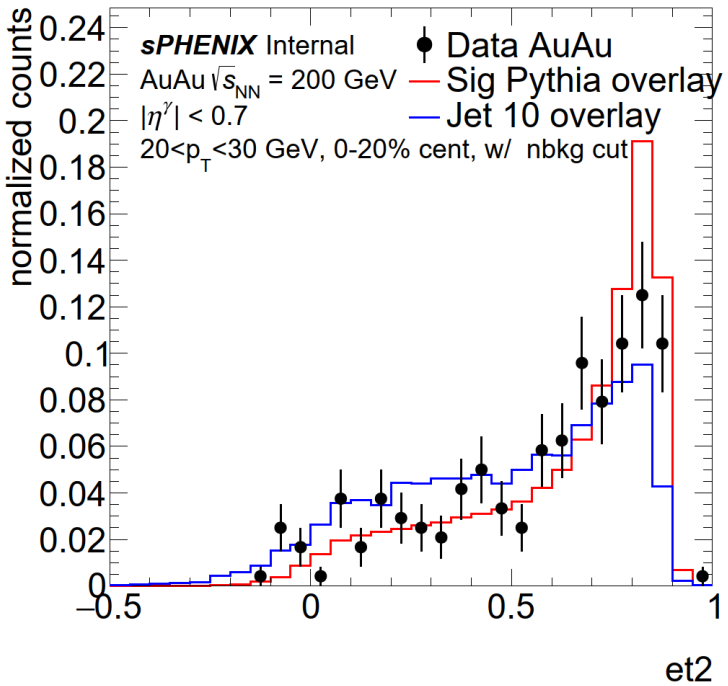


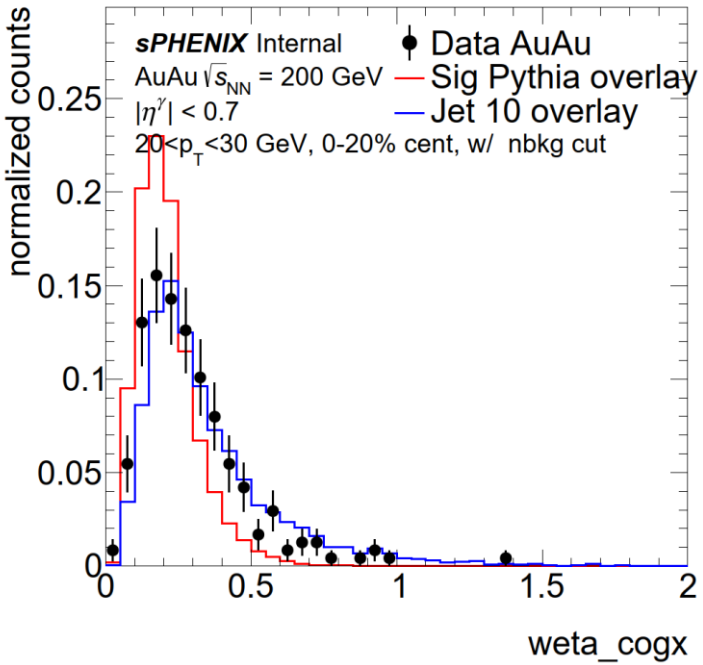


Shower shapes with NPB removed

- After NPB removal we can compare shower shapes data-MC to develop photon ID
- Jet overlay includes signal+bkg so in pp we expect it to grow with data, in AuAu we expect the bkg fraction to be overestimated.

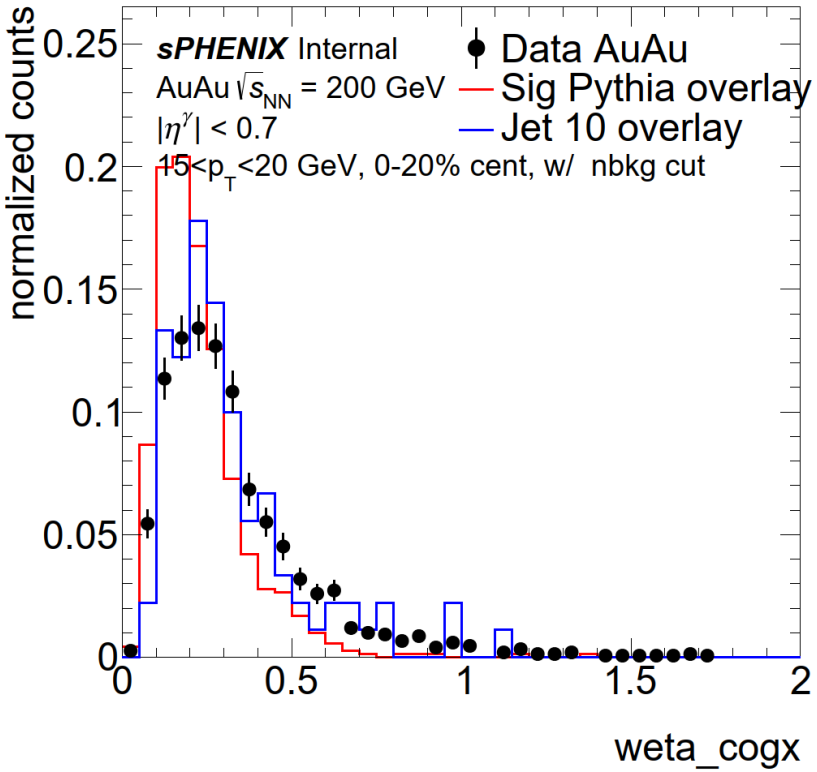


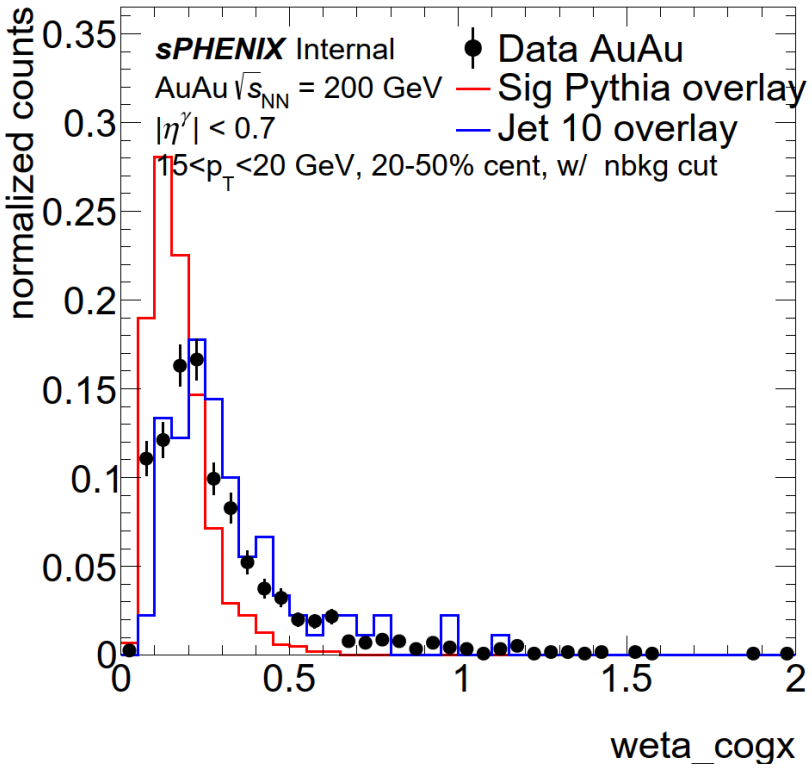


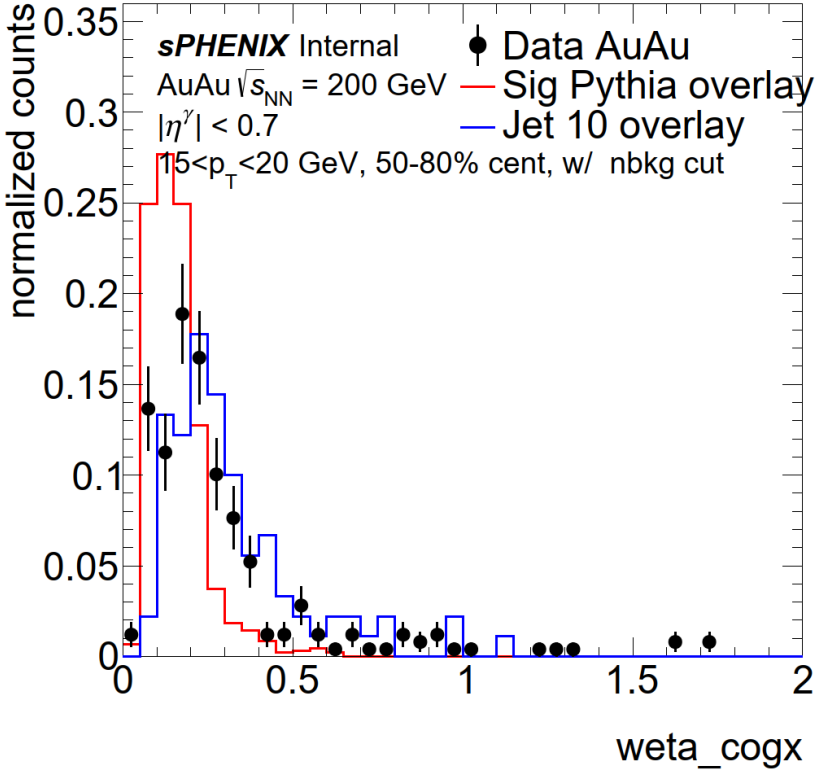


Shower shape evolution with centrality

- Trends in the width of weta_cogx is reproduced in the MC but is hard to fully validate at this point
- Seems possible that the photon ID will have to be centrality dependent
- Trying to find equivalent shower shapes that are less sensitive to UE

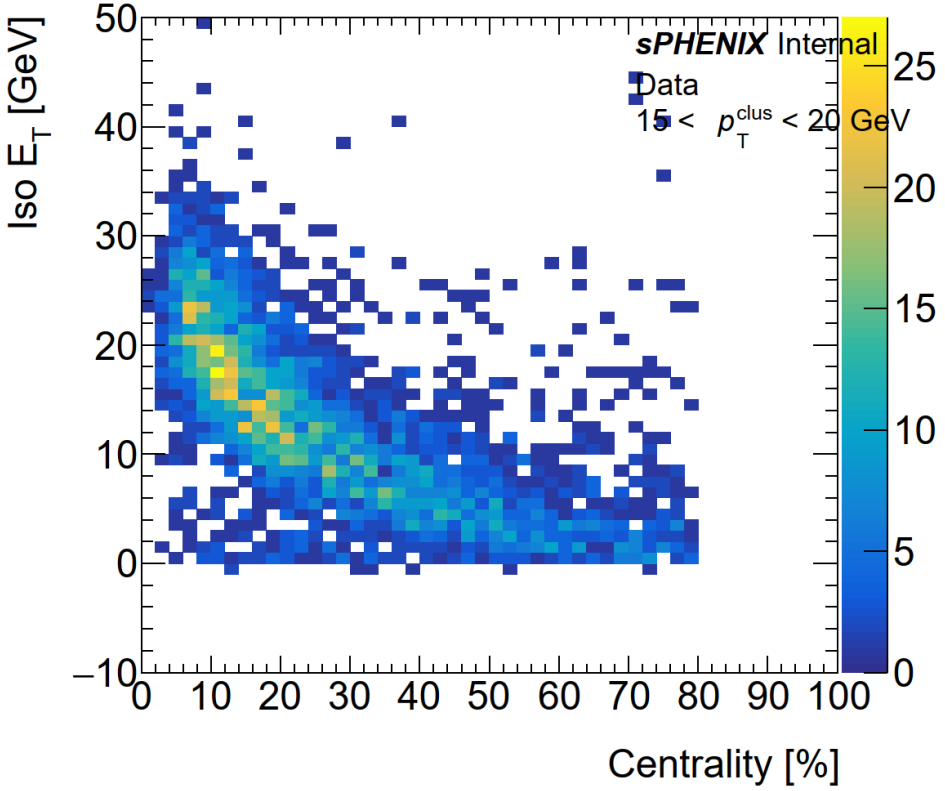


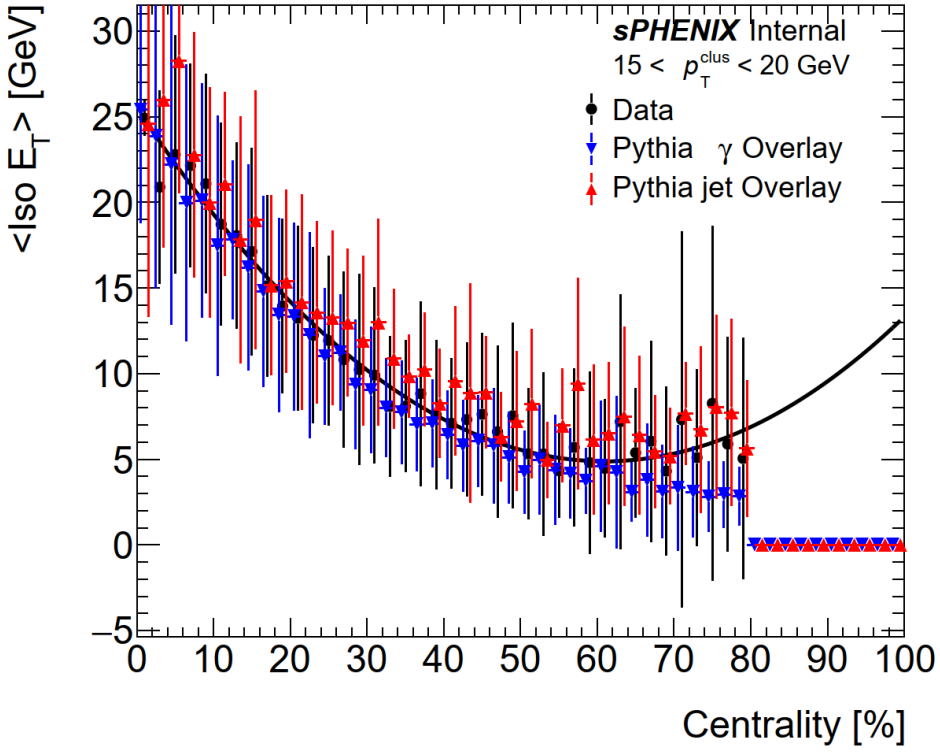




Isolation ET (no subtraction)

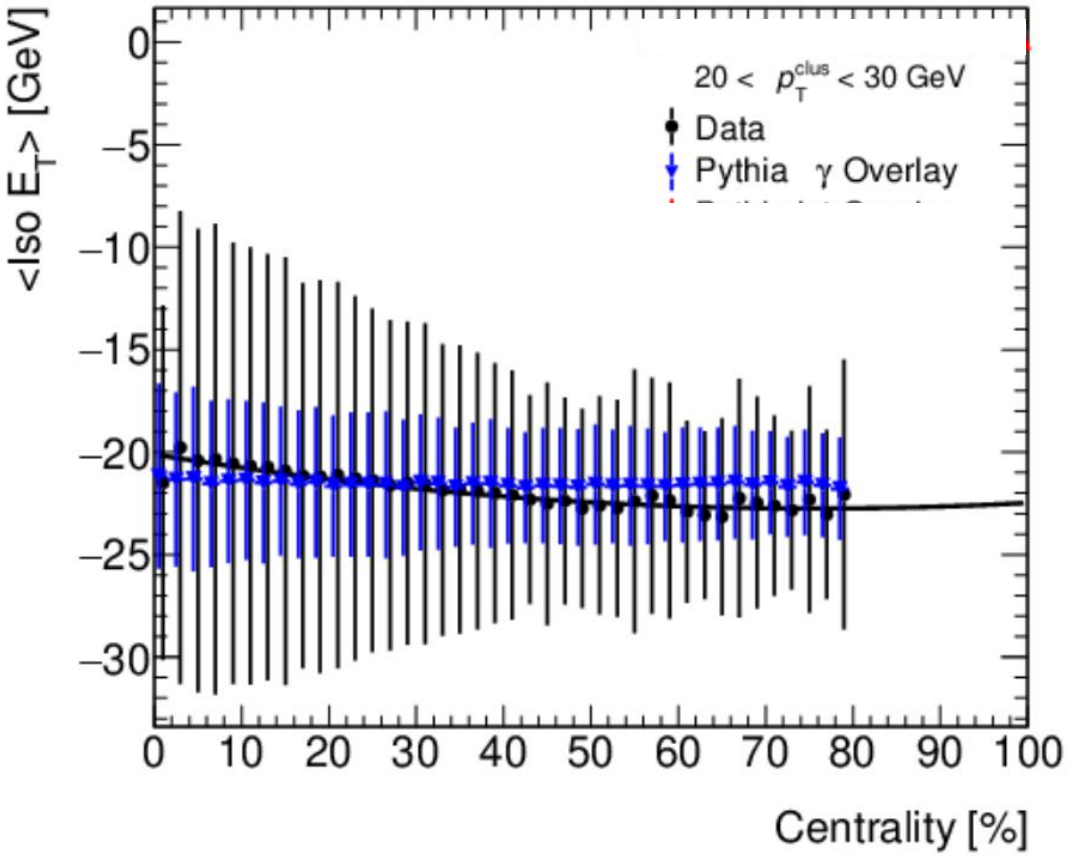
- Sum of ∑ET of all tower > 70 MeV
- Need to investigate low ET events in central events
-





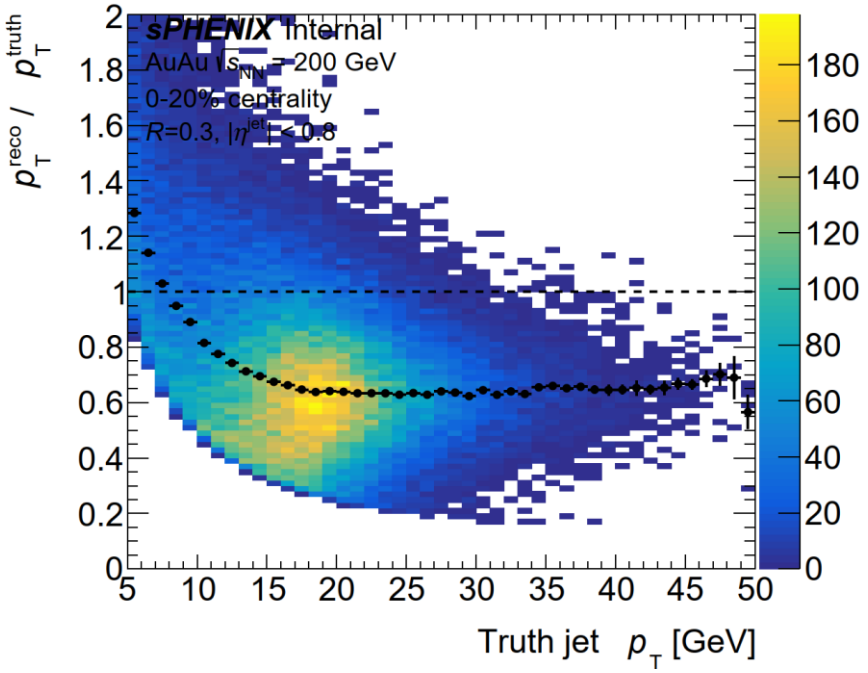
Isolation ET (with subtraction?)

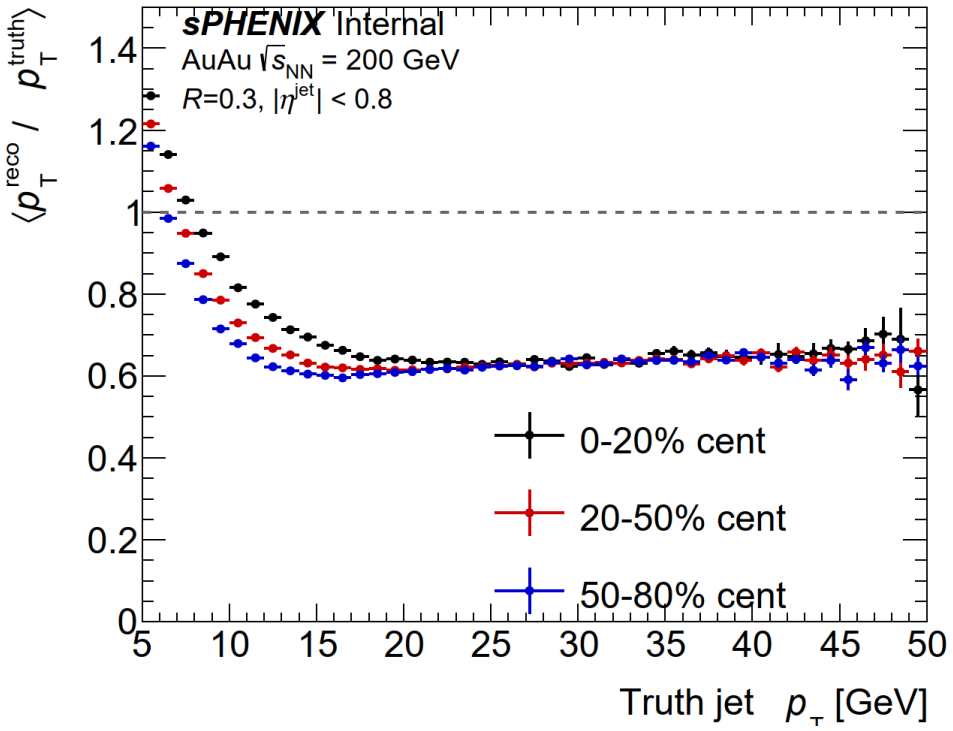
- **UE subtracted isolation ET**
  - Utilizing TOWERINFO_CALIB_CEMC_RETOWER_SUB1
  - Subtract cluster ET
- **Both look very wrong and the data and MC d not agree in width**
- **Need to investigate**

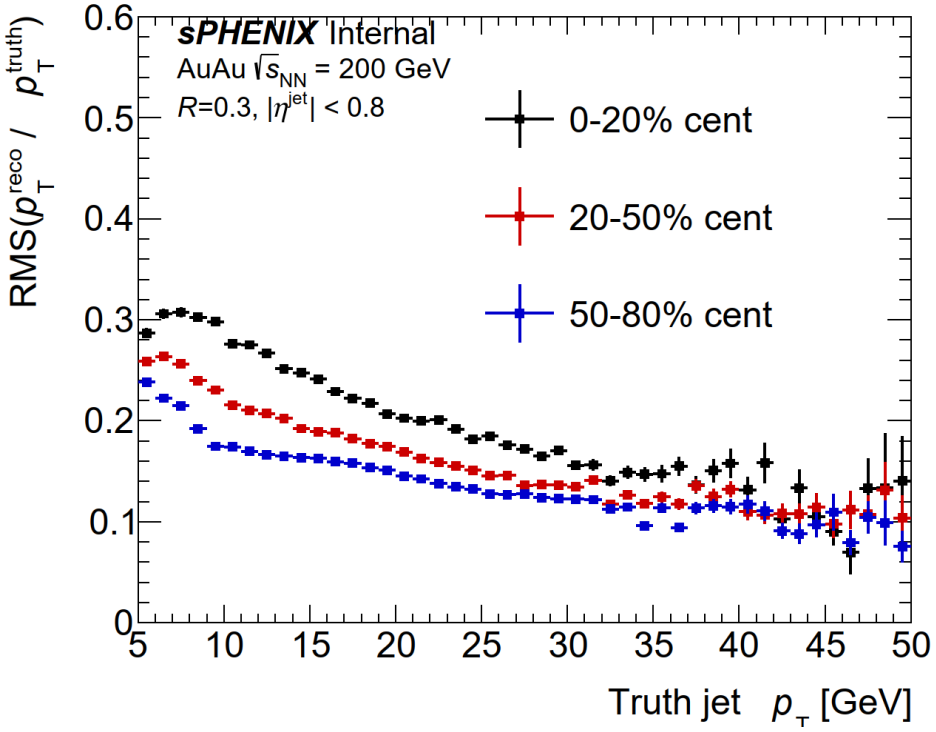


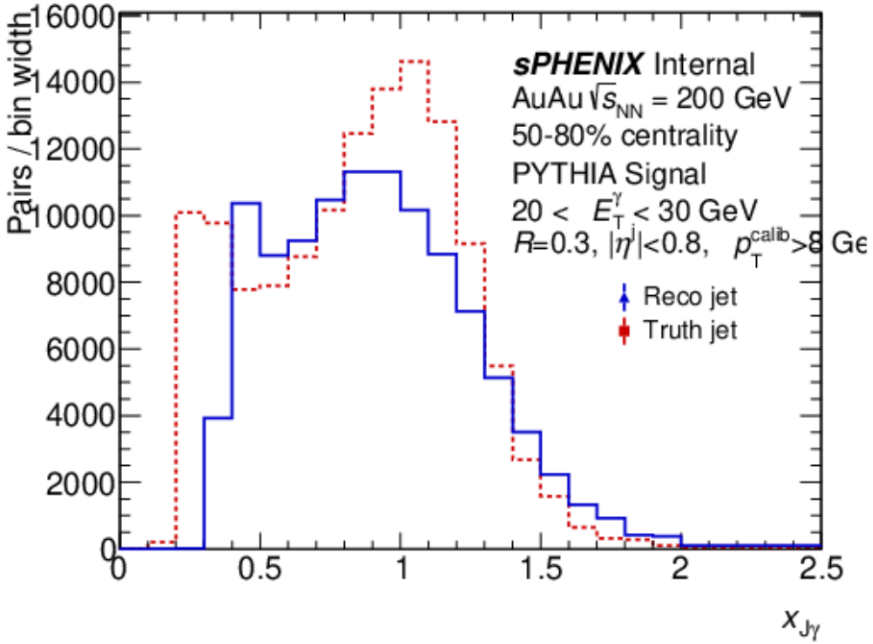
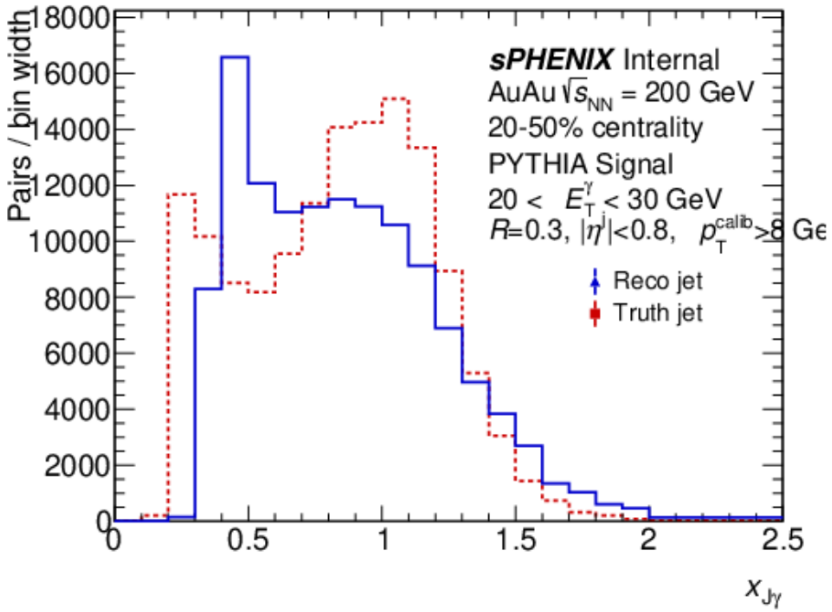
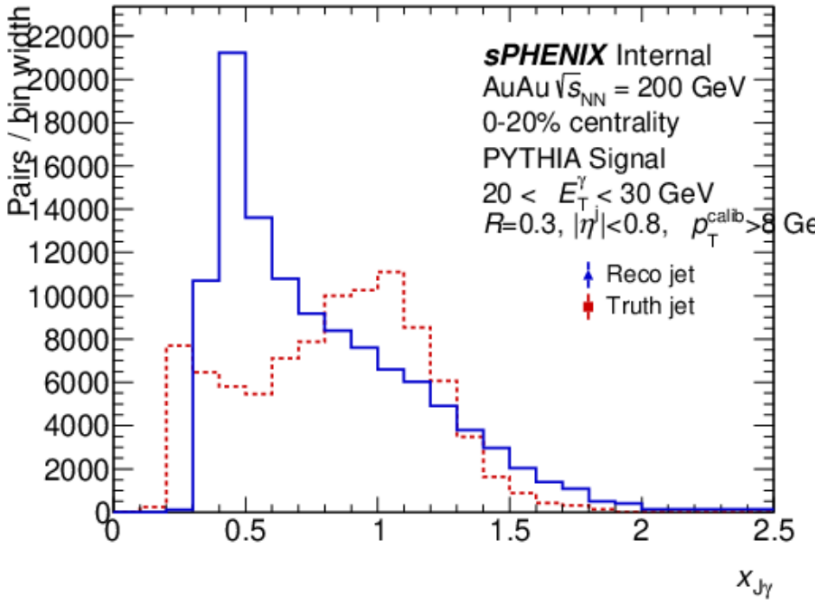
Check jet properties in

- **Using photon20 sample**
- **Left: response of jets**
  - Excluding the photon
  - Truth-reco ΔR < 0.2
  - Only saving uncalibrated jets > 5 GeV..
- **Middle and right: Kinda JES and JER**
  - Underlying subtraction seems to be behaving
  - Clear kinematic bias below ~15 GeV







$x_{j\gamma}$ in photon 20 sample

- **Construct reco level $x_{j\gamma}$ in photon 20 and separately with the truth jet (> 5 GeV)**
- **Include all jets back 7pi/8**
- **No peak in central collisions**
  - Is this poor JER or fake contribution
  - Need to do fake subtraction to find out